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### Communication apparatus establishing wireless communication using public key

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#### Abstract

A first terminal obtains a character string by decoding a code image. The character string is described in accordance with Bootstrapping Information Format and includes a public key of a communication apparatus and specific information. The first terminal displays an operation screen by using the specific information. The first terminal executes a first target communication with the communication apparatus by using the public key after a changing operation indicated by the operation screen is performed to the communication apparatus. A second terminal obtains the character string by decoding the code image. The second terminal executes a second target communication with the communication apparatus by using the public key after the changing operation is performed to the communication apparatus without displaying the operation screen using the specific information.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. Ser. No. 17/679,181 filed on Feb. 24, 2022 and claims priority from Japanese Patent Application No. 2021-029846 filed on Feb. 26, 2021, the entire contents of which are incorporated herein by reference.

### TECHNICAL FIELD

(1) Aspects disclosed herein relate to a technique for establishing a wireless connection between a pair of apparatuses in accordance with a Device Provisioning Protocol scheme (hereinafter referred to as a “DPP scheme”).

### BACKGROUND

(2) A DPP scheme that is a wireless communication scheme defined by Wi-Fi Alliance is known. The DPP scheme is for establishing a wireless connection according to Wi-Fi (Registered Trademark, Wi-Fi Alliance) standard between a pair of apparatuses.

### SUMMARY

(3) Aspects of the disclosure provide a technique that enables establishment of a wireless connection between a pair of apparatuses using the terminal easily according to the DPP scheme of the Wi-Fi standard.

(4) In an aspect of the disclosure, a communication system may include a communication apparatus, a first terminal, and a second terminal. The first terminal includes: a first camera; a first wireless interface configured to execute a wireless communication according to Device Provisioning Protocol (DPP) scheme of Wi-Fi standard; a display; a first computer; a first memory configured to store a first computer program that is provided by a vendor of the communication apparatus, wherein the first computer program, when executed by the first computer, causes the first terminal to: obtain a character string by decoding a code image in response to the first camera reading the code image, the character string being described in accordance with Bootstrapping Information Format and including a public key of the communication apparatus and specific information, the specific information being used for displaying an operation screen for the communication apparatus; execute a display process by using the specific information in the character string, the display process being for displaying the operation screen on the display, the operation screen indicating a changing operation for changing a state of the communication apparatus from a first state in which the wireless communication according to the DPP scheme is

not to be executed to a second state in which the wireless communication according to the DPP scheme is to be executed; and execute a specific process after the character string is obtained, the specific process being for enabling a first target communication via the first wireless interface with the communication apparatus according to the DPP scheme, the first target communication being for enabling a pair of apparatuses to establish a wireless connection therebetween according to the DPP scheme, and the first target communication being executed using the public key included in the character string after the state of the communication apparatus is changed from the first state to the second state due to the changing operation, wherein the second terminal comprises: a second camera; a second wireless interface configured to execute the wireless communication according to the DPP scheme; a second computer; and a second memory configured not to store the first computer program but to store a second computer program, the second computer program being not provided by the vendor of the communication apparatus, wherein the second computer program, when executed by the second computer, causes the second terminal to: obtain the character string by decoding the code image in response to the second camera reading the code image, the character string being described in accordance with Bootstrapping Information Format and including the public key of the communication apparatus and specific information, the specific information being used for displaying the operation screen for the communication apparatus; and execute, without executing the display process, the specific process after the character string is obtained, the specific process being for enabling a second target communication via the second wireless interface with the communication apparatus according to the DPP scheme, the second target communication being for enabling a pair of apparatuses to establish a wireless connection therebetween according to the DPP scheme, and the second target communication being executed using the public key included in the character string after the state of the communication apparatus is changed from the first state to the second state due to the changing operation.

(5) According to the above configuration, a user of the first terminal may perform the changing operation to the communication apparatus easily in accordance with the operation screen. That is, the first terminal may provide the user of the first terminal with the operation screen that enables the user to perform the changing operation to the communication apparatus easily. Thus, the user may establish a wireless connection easily according to the DPP scheme between the pair of apparatuses using the first terminal. In addition, according to the above configuration, the character string obtained from the code image is described in accordance with the Bootstrapping Information Format. Accordingly, the second terminal that does not include the first computer program may also use the public key included in the character string. Thus, the user may establish a wireless connection easily according to the DPP scheme between the pair of apparatuses using the second terminal.

(6) Another aspect of the disclosure is a non-transitory computer readable recording medium for storing a first computer program for a terminal. The terminal includes: a camera; a wireless interface configured to execute a wireless communication according to Device Provisioning Protocol (DPP) scheme of Wi-Fi standard; a display; and a computer. The first computer program, when executed by the computer, causes the terminal to: obtain a character string by decoding a code image in response to the camera reading the code image, the character string being described in accordance with Bootstrapping Information Format and including a public key of a communication apparatus and specific information, the specific information being used for displaying an operation screen for the communication apparatus; execute a display process by using the specific information in the character string, the display process being for displaying the operation screen on the display, the operation screen indicating a changing operation for changing a state of the communication apparatus from a first state in which a wireless communication according to the DPP scheme is not to be executed to a second state in which the wireless communication according to the DPP scheme is to be executed; and execute a specific process after the character string is obtained, the specific process being for enabling a first target communication

via the wireless interface with the communication apparatus according to the DPP scheme, the first target communication being for enabling a pair of apparatuses to establish a wireless connection therebetween according to the DPP scheme, and the first target communication being executed using the public key included in the character string after the state of the communication apparatus is changed from the first state to the second state due to the changing operation.

(7) According to the above configuration, a user of the terminal may perform the changing operation to the communication apparatus easily in accordance with the operation screen. That is, the terminal may provide the user of the terminal with the operation screen that enables the user to perform the changing operation to the communication apparatus easily. Thus, the user may establish a wireless connection easily according to the DPP scheme between the pair of apparatuses using the terminal.

(8) Another aspect of the disclosure is a communication apparatus including: a wireless interface configured to execute a wireless communication according to Device Provisioning Protocol (DPP) scheme of Wi-Fi standard; a housing to which a seal is attached, the seal including a code image that is generated by encoding a character string, the character string being described in accordance with Bootstrapping Information Format and including a public key of the communication apparatus and specific information, the specific information being used for enabling a first terminal to execute a display process for displaying an operation screen of the communication apparatus, the display process being executed by a first computer program of the first terminal provided by a vender of the communication apparatus, the operation screen indicating a changing operation for changing a state of the communication apparatus from a first state in which a wireless communication according to the DPP scheme is not to be executed to a second state in which the wireless communication according to the DPP scheme is to be executed; and a controller configured to: receiving the changing operation for changing the state of the communication apparatus from the first state to the second state; in a case where the changing operation is received in response to displaying the operation screen at the first terminal, execute a first target communication via the wireless interface with the first terminal according to the DPP scheme in response to the first terminal using the public key included in the character string, the first target communication being for enabling a pair of apparatuses to establish a wireless connection therebetween according to the DPP scheme; and in a case where the changing operation is received without displaying the operation screen at a second terminal that stores a second computer program not provided by the vender, execute a second target communication via the wireless interface with the second terminal according to the DPP scheme in response to the second computer program using the public key included in the character string, the second target communication being for enabling a pair of apparatuses to establish a wireless connection therebetween according to the DPP scheme.

(9) According to the above configuration, a user of the first terminal may perform the changing operation to the communication apparatus easily in accordance with the operation screen. That is, the first terminal may provide the user of the first terminal with the operation screen that enables the user to perform the changing operation to the communication apparatus easily. Thus, the user may establish a wireless connection easily according to the DPP scheme between the pair of apparatuses using the first terminal. In addition, according to the above configuration, the character string obtained from the code image is described in accordance with the Bootstrapping Information Format. Accordingly, the second terminal including the second computer program that is not provided by a vendor of the communication apparatus may also use the public key included in the character string. Thus, the user may establish a wireless connection easily according to the DPP scheme between the pair of apparatuses using the second terminal.

(10) The terminal and the method performed by the terminal are also novel and useful. A non-transitory computer-readable recording medium storing the first computer program for the terminal is also novel and useful. Further, a method executed by the communication apparatus, a computer

program for realizing operations of the communication apparatus, and a computer-readable recording medium storing the computer program are also novel and useful.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a schematic configuration of a communication system.
- (2) FIG. 2 illustrates a control configuration of each apparatus.
- (3) FIG. 3 illustrates a schematic sequence diagram of processes for establishing a Wi-Fi connection between a printer and an access point.
- (4) FIG. 4 illustrates a sequence diagram of Bootstrapping of case A according to the first embodiment.
- (5) FIG. 5 illustrates a sequence diagram of Bootstrapping of case B according to the first embodiment.
- (6) FIG. 6 illustrates a sequence diagram of Bootstrapping according to the second embodiment.

### DETAILED DESCRIPTION

#### First Embodiment

- (7) (Configuration of Communication System 2; FIG. 1)

(8) As illustrated in FIG. 1, the communication system 2 includes an access point (hereinafter, referred to as “AP”) 6, a plurality of terminals 10A and 10B, a printer 100, and a server 200. In this embodiment, a situation is assumed in which a user uses one of the terminals 10A and 10B to establish a wireless connection (hereinafter referred to as a “Wi-Fi connection”) according to the Wi-Fi standard between the printer 100 and the AP 6.

- (9) (Configuration of Terminal 10A; FIG. 2)

(10) The terminal 10A is a portable terminal such as a mobile phone (e.g., a smartphone), a PDA, or a tablet PC. In a modification, the terminal 10A may be a stationary terminal. As illustrated in FIG. 2, the terminal 10A includes an operation unit 12, a display 14, a Wi-Fi interface 16, a camera 20, and a controller 30. Each unit 12 to 30 is connected to a bus (reference numeral omitted).

Hereinafter, the interface is referred to as an “I/F”.

(11) The operation unit 12 includes a plurality of buttons. The user may operate the operation unit 12 to input various instructions into the terminal 10A. The display 14 is for displaying various items of information. The display 14 includes a touchscreen function (i.e., an operation unit). The camera 20 is for reading an image of an object. In this embodiment, the camera 20 is used to read a QR Code (Registered Trademark, Denso Wave Incorporated Corporation) in association with the printer 100.

(12) The Wi-Fi I/F 16 is a wireless interface for performing wireless communication in accordance with the Wi-Fi standard. The Wi-Fi standard is for performing wireless communication in accordance with a standard such as the 802.11 standard of Institute of Electrical and Electronics Engineers, Inc. (IEEE) and similar specifications modeled after IEEE 802.11 (e.g., 802.11a, 11b, 11g, 11n). The terminal 10A may establish a Wi-Fi connection with AP 6 through Wi-Fi I/F 16 by using an SSID and a password of a wireless network formed by AP 6.

(13) The Wi-Fi I/F 16 supports the DPP scheme established by Wi-Fi Alliance. The DPP scheme is described in a standard document of “Wi-Fi Easy Connect Specification Version 2.0” created by the Wi-Fi Alliance, and is for easily establishing a Wi-Fi connection between a pair of apparatuses (e.g., the printer 100 and the AP 6) using a terminal 10A. Hereinafter, this standard document is referred to as “DPP standard document”.

(14) The controller 30 includes a CPU 32 and a memory 34. The CPU 32 executes various processing in accordance with each program 40, 41, and 42 stored in the memory 34. The memory 34 is a memory such as a volatile memory and a nonvolatile memory. The memory 34 stores an OS

program **40**, a connection application **41**, and a printer application **42**. Hereinafter, the OS program **40** is referred to as “OS **40**”.

(15) The OS **40** is a program for causing basic operations of the terminal **10A**. In this embodiment, the OS **40** is Android (Registered trademark, Google, Inc). In modification, the OS **40** may be another OS (e.g., iOS (Registered Trademark, Cisco systems, Inc)). The OS **40** is prestored in the terminal **10** at the shipment of the terminal **10A**.

(16) The connection application **41** is a program for causing a wireless communication according to the DPP scheme to establish a Wi-Fi connection between the printer **100** and the AP **6**. The connection application **41** is a standard application used together with the OS **40**. The connection application **41** is stored in advance in the terminal **10A** before shipment of the terminal **10A**. That is, the connection application **41** is a program that is not provided by the vendor of the printer **100**. In a modification, the terminal **10A** may include a universal application instead of the connection application **41**.

(17) The printer application **42** is a program for causing the connection application **41** to execute a wireless communication in accordance with the DPP scheme. The printer application **42** is a program for causing the printer **100** to execute printing. The printer application **42** is a program provided by the vendor of the printer **100**. The printer application **42** is installed in the terminal **10A** after the shipment of the terminal **10A**.

(18) (Configuration of Terminal **10B**; FIG. 2)

(19) The terminal **10B** basically has the same configuration as that of the terminal **10A**. That is, the terminal **10B** includes an operation unit, a display, a Wi-Fi I/F, a camera, and a controller including a CPU and a memory (not shown in figures). The memory of the terminal **10B** stores an OS **50** and a connection application **52**, and does not store the printer application **42**. The connection application **52** is the same application as the connection application **41** in the terminal **10A**. The connection application **52** is prestored in the terminal **10B** at the shipment of the terminal **10B**.

(20) (Configuration of Printer **100**; FIGS. 1 and 2)

(21) The printer **100** is a peripheral device (e.g., a peripheral device of terminals **10A** and **10B**) configured to perform printing. The printer **100** has a model name “AAA” and a serial number “xyz”. As illustrated in FIG. 1, the printer **100** includes a housing **100A**. The housing **100A** has a seal **100B** indicating a QR Code attached thereto. The QR Code is information obtained by encoding a character string including a Uniform Resource Identifier (URI) and a public key of the printer **100**. The URI includes a domain of the server **200** and the serial number “xyz” of the printer **100**.

(22) As illustrated in FIG. 2, the printer **100** includes an operation unit **112**, a Wi-Fi I/F **116**, a print execution unit **118**, and a controller **130**. Each unit **112** to **130** is connected to a bus line (reference numeral omitted). The printer **100** is not configured to display a QR Code. This means that the printer **100** does not have any display or that the printer **100** has a very small display.

(23) The operation unit **112** includes a plurality of buttons. The user may input various instructions to the printer **100** using the operation unit **112**. The Wi-Fi I/F **116** supports the DPP scheme. The print execution unit **118** includes a print mechanism such as an inkjet system or a laser system. The controller **130** includes a CPU **132** and a memory **134**. The CPU **132** executes various processing in accordance with a program **136** stored in the memory **134**. The memory **134** is a memory such as a volatile memory and a nonvolatile memory.

(24) (Configuration of Server **200**; FIG. 2)

(25) The server **200** is a web server, and is disposed on the Internet **4** (illustrated in FIG. 1) by a vendor of the printer **100**. The server **200** includes a communication I/F **216** and a controller **230**. Each unit **216** and **230** is connected to a bus (reference numeral omitted). The communication I/F **216** is connected to the Internet **4**. The controller **230** includes a CPU **232** and a memory **234**. The CPU **132** executes various processing in accordance with a program **236** stored in the memory **234**. The memory **234** is a memory such as a volatile memory and a nonvolatile memory.

(26) The memory **234** further stores a model table **250**. In the model table **250**, for each of a plurality of model names of a printer, the model name is associated with a message indicating a changing operation. The changing operation is for changing the state of the printer having the model name from an OFF state to an ON state. The OFF state is a state in which a wireless communication according to the DPP scheme is not to be executed. The ON state is a state in which the wireless communication according to the DPP scheme is to be executed.

(27) (Overview of DPP; FIG. 3)

(28) Hereinafter, an overview of the DPP will be described with reference to FIG. 3. In the following, for ease of understanding, an operation executed by a CPU (e.g., a CPU **32**) of each apparatus is mainly described by each apparatus (e.g., a terminal **10A**) without being mainly described by the CPU. In addition, each of the terminal **10A** and the terminal **10B** executes the same processing in FIG. 3. Thus, in FIG. 3, the terminal **10A** and the terminal **10B** are collectively referred to as “terminal **10**”. The following communications are executed by using I/F **16**, **116**, and **216** (illustrated in FIG. 2). Thus, the description of “via I/F” is omitted.

(29) In this embodiment, the AP **6** does not support the DPP scheme. The AP **6** operates as a parent station of the wireless network in which an SSID of “sss” and a password of “ttt” are used. For example, the user of the terminal **10** inputs the SSID of “sss” and the password of “ttt” to the terminal **10** by operating the operation unit **12**. In response to this operation, the terminal **10** establishes a Wi-Fi connection with the AP **6** in accordance with the OS **40**. The terminal **10** may communicate with the printer **100** via the AP **6** to cause the printer **100** to execute printing after the printer **100** has established a Wi-Fi connection with the AP **6**. This is achieved by the processing described below.

(30) In T25, the terminal **10** executes Bootstrapping (hereinafter, referred to as “BS”) with the printer **100**. The BS is described in the DPP standard document. The BS is a process in which the printer **100** provides the terminal **10** with a public key in response to the terminal **10** reading a QR Code indicated in a seal **100B** (as illustrated in FIG. 1) attached to the printer **100**. The public key is used in an Authentication (hereinafter referred to as “Auth”) described in the DPP standard document.

(31) In T30, the terminal **10** executes Auth with the printer **100** using the public key obtained through the BS in T25. Auth is a process for each of the terminal **10** and the printer **100** to authenticate a communication partner. Specifically, the terminal **10** transmits an Auth request using the public key to the printer **100**, and receives an Auth response from the printer **100**.

(32) In T35, the terminal **10** executes Configuration (hereinafter referred to as “Config”) with the printer **100**. The Config is described in the DPP standard document. The Config is a process of transmitting, to the printer **100**, information for establishing a Wi-Fi connection between the printer **100** and the AP **6**. Specifically, the terminal **10** generates a Signed Connector (hereinafter referred to as “SC”) and transmits the SC to the printer **100**. The SC includes the SSID of “sss” of AP **6** and the password of “ttt” of AP **6**.

(33) In T40, the printer **100** establishes a Wi-Fi connection with AP **6** using the SSID of “sss” and the password of “ttt” included in the SC obtained in T35. Specifically, the printer **100** transmits a Probe request by broadcast. Then, the printer **100** receives a Probe response from each of one or more APs around the printer **100** including the AP **6**. In this case, the printer **100** determines that received one or more Probe responses include a Probe response including the SSID of “sss” that is the same SSID included in the received SC. Then, the printer **100** executes various communications such as an Association and a 4way-handshake with the AP **6**, thereby establishing a Wi-Fi connection with the AP **6**. In the process of the various communications, the printer **100** transmits authentication information using the password of “ttt” included in the received SC to the AP **6**, and the AP **6** performs authentication of the password of “ttt”. In response to a success of the authentication, a Wi-Fi connection is established between the printer **100** and the AP **6**.

(34) As a result of the processes illustrated in FIG. 3, both of the terminal **10** and the printer **100**



belongs to the same wireless network formed by AP 6 as a child station. Accordingly, the terminal 10 may transmit print data to the printer 100 via the AP 6. Then, the printer 100 can print an image represented by the print data.

(35) In the DPP scheme, input of the SSID of “sss” of the AP 6 and the password of “ttt” of the AP 6 to the printer 100 in order to establish the Wi-Fi connection between the printer 100 and the AP 6 may be omitted. Thus, the Wi-Fi connection between the printer 100 and the AP 6 may be established easily.

(36) (Details of BS; FIGS. 4 and 5)

(37) Referring to FIGS. 4 and 5, the BS in the T25 of FIG. 3 will be described in detail. FIG. 4 illustrates the BS of the case A executed by the terminal 10A having the printer application 42. FIG. 5 illustrates the BS of the case B executed by the terminal 10B not having the printer application 42.

(38) (Case A; FIG. 4)

(39) The QR Code represented by the seal 100B of the printer 100 is a coded character string 150. The character string 150 is described in accordance with the Bootstrapping Information Format described in the DPP standard document. Specifically, the character string 150 includes an identifier “K” according to an Augmented Backus Naur Form (ABNF) rule, a character string “12345abcd” described in association with the identifier “K”, an identifier “I” according to the ABNF rule, and a character string “MAN=https://www.server.com/xyz” described in association with the identifier “I”. In this embodiment, “a character string described in association with an identifier” means a character string described after the identifier.

(40) The identifier of “K” indicates that the character string of “12345abcd” associated with the identifier “K” is the public key of the printer 100. The connection applications 41 and 52, which are programs supporting the DPP scheme, may recognize that the character string specified by the identifier of “K” is a public key. Thus, the connection applications 41 and 52 may appropriately specify the public key of the printer 100.

(41) The identifier “I” indicates that the character string of “MAN=https://www.server.com/xyz” associated with the identifier “I” is information described by the vendor of the printer 100 arbitrarily. In this embodiment, the information corresponds to “MAN” that indicates a manual (i.e., an operation screen) in the server 200 and “https://www.server.com/xyz” that is a URI of the manual. The URI includes “xyz” that is the serial number of the printer 100. The printer application 42, which is a program provided by the vendor of the printer 100, may recognize that the character string specified by the identifier of “I” is a URI and that the URI includes the serial number of the printer 100. The connection applications 41 and 52, which are programs that are not provided by the vendor of the printer 100, may not recognize that the character string specified by the identifier of “I” is the URI of the server 200. In other words, the character string specified by the identifier of “I” is configured such that the printer application 42 may recognize that the character string is the URI of the server 200 and the connection applications 41 and 52 may not recognize that the character string is the URI of the server 200. Thus, the printer application 42 may appropriately specify the URI of the server 200.

(42) In T100, the terminal 10A receives an operation to activate the printer application 42 from the user. The terminal 10A executes the following processing in accordance with the printer application 42. In T110, in response to receiving, from the user, an operation of reading the QR Code indicated by the seal 100B, the terminal 10A reads the QR Code with the camera 20. In T120, the terminal 10A obtains the character string 150 by decoding the QR Code. In T122, the terminal 10A specifies the URI 160 of “https://www.server.com/xyz” specified by the identifier of “I” from the character string 150.

(43) In T130, the terminal 10A transmits, to the web server 200, a web page request including URI 160 that is a transmission destination. Although the terminal 10A includes a browser program (not shown in figures), the terminal 10A transmits the web page request and displays a web page using

the printer application **42** without using the browser program. According to this configuration, the terminal **10A** does not need to activate the browser program. In other words, the terminal **10A** does not need to display a window displayed by the browser program that is different from the window displayed by the printer application **42**. Thus, the processing load on the terminal **10A** may reduce, and the user does not need to view a plurality of windows.

(44) In response to receiving the web page request from the terminal **10A** in **T130**, in **T132**, the server **200** specifies the serial number of “xyz” included in the URI **160** of the web page request. The server **200** stores a serial number table (not shown in figures). The table includes, for each of a plurality of printers, a serial number of the printer and a model name of the printer in association with each other. Thus, in **T132**, the server **200** may identify the model name of “AAA” of the printer **100** from the serial number of “xyz” by using the table. In modification, the serial number of “xyz” may include a character string indicating the model name of “AAA”. In this modification, the server **200** may specify the model name of “AAA” from the serial number “xyz” without using the serial number table.

(45) In **T134**, the server **200** extracts a message of “Press the Wi-Fi button three times” corresponding to the model name “AAA” specified in **T132** from the model table **250** (illustrated in FIG. 2) in the memory **234**. Next, the server **200** generates web page data representing a web page. The web page includes a message **302** including the extracted message and a predetermined message, and a start button. The predetermined message indicates that the start button needs to be pressed after the operation according to the extracted message is performed. The start button is a button for causing the terminal **10A** to start Auth. Then, in **T136**, the server **200** transmits the generated web page data to the terminal **10A**.

(46) In response to receiving the web page data from the server **200** in **T136**, in **T140**, the terminal **10A** displays the Web page **300** represented by the Web page data on the display **14**. The web page **300** includes the message **302** and a start button.

(47) In **T150**, the printer **100** determines that the user has been pressed the Wi-Fi button included in the operation unit **112** of the printer **100** based on the message **302** in the web page **300**. In response to this determination, in **T152**, the printer **100** shifts the status of the printer **100** from the OFF state to the ON state. More specifically, the printer **100** (i.e., CPU **132**) provides, to the Wi-Fi I/F **116**, an instruction of switching from the OFF state to the ON state. Accordingly, the printer **100** is enabled to communicate according to the DPP scheme (e.g., to receive an Auth request or to transmit an Auth response). That is, the web page **300** is an operation screen displaying a changing operation for changing the state of the printer **100** from the OFF state to the ON state.

(48) According to this embodiment, the web page **300** includes the message **302** for changing the state of the printer **100** indicated by the model name of “AAA” to the ON state. Thus, the printer **100** may change the state of the printer **100** to the ON state easily in response to the user's operation. If the printer **100** maintains the state of the printer **100** in the OFF state, the printer **100** does not transmit the Auth response to the terminal **10A**. In this case, the Auth is not executed, and the printer **100** does not establish the Wi-Fi connection with the AP **6**.

(49) In **T160**, the terminal **10A** receives a user's operation of selecting the start button in the web page **300**. In response to the user's operation, in **T162**, the terminal **10A** (i.e., the printer application **42**) provides the character string **150** specified in **T120** to the connection application **41**.

(50) In **T170**, the terminal **10A** (i.e., the connection application **41**) specifies the public key **170** indicated by the identifier of “K” from the character string **150**. Thereafter, the terminal **10A** (i.e., the connection application **41**) executes the processing of the steps **T30** and **T35** of FIG. 3. That is, the terminal **10A** transmits an Auth request using the public key **170** specified in **T170** to the printer **100**. In other words, the terminal **10A** does not transmit the Auth request to the printer **100** until the start button is operated. Thus, while the printer **100** is in the OFF state, the transmission of the Auth request from the terminal **10A** to the printer **100** is not performed.

(51) As described with reference to FIG. 3, the terminal **10A** transmits the Auth request to the

printer **100**, receives the Auth response from the printer **100** (i.e., T30 of FIG. 3), and transmits the SC to the printer **100** (i.e., T35 of FIG. 3). Thus, the terminal **10A** may establish the Wi-Fi connection with the AP **6** (i.e., T40 of FIG. 3).

(52) (Case B; FIG. 5)

(53) Hereinafter, BS in the case B executed by the terminal **10B** will be described. In T200, the terminal **10B** receives an operation to activate the connection application **52** from the user. Then, the terminal **10B** executes the following processing in accordance with the connection application **52**. In T210, in response to receiving, from the user, an operation of reading the QR Code indicated by the seal **100B**, the terminal **10B** reads the QR Code with a camera of the terminal **10B**. In T220, the terminal **10B** obtains the character string **150** by decoding the QR Code. In T222, the terminal **10B** specifies the public key indicated by the identifier of “K” from the character string **150**. In case B, the terminal **10B** may not specify the URI **160** indicated by the identifier “I”.

(54) Accordingly, the terminal **10B** does not execute the processing for displaying the web page **300** using the URI **160** (i.e., T130, T136, T140 in FIG. 4). In this embodiment, a situation is assumed in which the user of the terminal **10B** knows the operation for changing the state of the printer **100** to the ON state (i.e., pressing the Wi-Fi button three times) in advance. Thus, the user may press the Wi-Fi button three times in T250 without viewing the web page **300**. Accordingly, in T252, the printer **100** changes the state of the printer **100** from the OFF state to the ON state.

(55) In response to specifying the public key **170** in T222, the terminal **10B** repeatedly transmits the Auth request using the public key **170** to the printer **100** without waiting for the operation from the user (e.g., an operation of pressing a start button in T160 of FIG. 4). After transmitting the Auth request to the printer **100**, the terminal **10B** receives the Auth response from the printer **100** (i.e., T30 of FIG. 3), and then transmits the SC to the printer **100** (i.e., T35 of FIG. 3). Thus, the terminal **10B** may establish the Wi-Fi connection with the AP **6** (i.e., T40 of FIG. 3).

Effects of the Embodiment

(56) According to the above configuration, the user of the terminal **10A** may perform changing operation to the printer **100** in accordance with the displayed web page **300**. That is, the printer **100** may receive the changing operation from the user in accordance with the web page **300** displayed on the terminal **10A** (T40 of FIG. 4). Thus, the printer **100** may establish a Wi-Fi connection with the AP **6** easily according to the DPP scheme using the terminal **10A**. In addition, according to the above configuration, the character string **150** obtained from the code image is described in accordance with the Bootstrapping Information Format. Thus, the terminal **10B** not storing the printer application **42** may determine the public key **170** included in the character string **150**. Thus, the printer **100** may establish a Wi-Fi connection with the AP **6** according to the DPP scheme using the terminal **10B**.

(57) The character string **150** indicated by the QR Code is described in accordance with the Bootstrapping Information Format. Thus, the QR Code may pass the authentication test of the Wi-Fi Alliance. Thus, a terminal not storing the printer application **42** may use the QR Code of the printer **100**.

(58) (Correspondence Relationship)

(59) The printer **100** is an example of a “communication apparatus”. The terminal **10A** is an example of a “first terminal”. The terminal **10B** is an example of a “second terminal”. A pair of the printer **100** and the AP **6** is an example of a “pair of apparatuses”. The printer application **42** is an example of a “first computer program”. The connection application **52** is an example of a “second computer program”. The OFF state is an example of a “first state”. The ON state is an example of a “second state”. The message **302** in the web page **300** is an example of a “specific message”. The identifier of “K” is an example of a “first identifier”. The identifier of “I” is an example of a “second identifier”. The URI **160** is an example of “specific information”. The serial number of “xyz” is an example of “identification information”.

(60) The processing of T30 in FIG. 3 executed by the terminal **10A** is an example of “first target

communication". The processing of T35 in FIG. 3 executed by the terminal **10A** is an example of "target communication". The Auth request transmitted from the terminal **10A** to the printer **100** is an example of each of a "first authentication request" and an "authentication request". The Auth response transmitted from the printer **100** to the terminal **10A** is an example of each of a "first authentication response" and an "authentication response". The processing of T30 and T35 in FIG. 3 executed by the terminal **10B** is an example of a "second target communication". The Auth request transmitted from the terminal **10B** to the printer **100** is an example of a "second authentication request". The Auth response transmitted from the printer **100** to the terminal **10B** is an example of a "second authentication response". The SC transmitted from the terminal **10A** to the printer **100** is an example of each of "connection information" and "first connection information". The SC transmitted from the terminal **10B** to the printer **100** is an example of "second connection information".

#### Second Embodiment; FIG. 6

(61) Hereinafter, a second embodiment will be described. Description of the configuration common to the first embodiment will be omitted. As illustrated in FIG. 6, a character string **152** indicated by a QR Code in the second embodiment is different from the QR Code of the first embodiment. The character string **152** includes the character string of "12345abcd" described in association with the identifier of "K" that is the same as the character string of the first embodiment, and the character string of "SN=xyz" described in association with the identifier of "I" that is not the same as the character string of the first embodiment. The character string of "SN=xyz" includes a character string of "SN" indicating a serial number and a serial number of "xyz" of the printer **100**. The printer application **42**, which is a program provided by the vendor of the printer **100**, may determine that the character string specified by the identifier of "I" includes the serial number of "xyz". The connection applications **41** and **52** may not determine that the character string specified by the identifier of "I" includes the serial number of "xyz".

(62) In T300, the terminal **10A** receives an operation to activate the printer application **42** from the user. In T310, in response to receiving an operation of reading the QR Code indicated by the seal **100B**, the terminal **10A** reads the QR Code with the camera **20**. In T320, the terminal **10A** obtains the character string **152** by decoding the QR Code. The terminal **10A** in the second embodiment obtains the character string **152** instead of the character string **150** of the first embodiment. In T332, the terminal **10A** specifies the serial number **162** of "xyz" indicated by the identifier "I" from the character string **152**.

(63) In T324, the terminal **10A** generates URI **164** of "https://www.server.com/xyz". The URI **164** includes the domain of the server **200** of "www.server.com" that the printer application **42** can determine due to storing the domain therein in advance, and the serial number "xyz" specified in T322. According to this configuration that the terminal **10A** generates the URI **164**, the terminal **10A** may determine appropriate URI in a situation where a location of the web page in the server **200** may change. That is, in response to notifying the changed position to the printer application **42**, the printer application **42** may appropriately generate a URI indicating the changed location.

(64) Steps T330 to T370 are the same as the steps T130 to T170 in FIG. 4. Each process executed by the terminal **10B** and the printer **100** is the same as the steps T200 to T252 in FIG. 5. The serial number of "xyz" is an example of "specific information".

(65) Although specific examples of the technology disclosed in the specification have been described in detail above, these are merely examples and do not limit the scope of the claims. The technology described in the claims includes various modifications and changes of the specific examples illustrated above. Modifications are listed below.

#### First Modification

(66) The terminal **10** may establish the Wi-Fi connection with the AP **6** according to a wireless communication of DPP scheme, instead of using the SSID of "sss" and the password of "ttt" of the AP **6**. In this modification, the AP **6** supports the DPP scheme, and has the QR Code indicating a

public key of the AP **6** (e.g., has a seal attached to a housing thereof). The terminal **10** obtains the public key of the AP **6** by reading the QR Code (i.e., by executing BS). Thereafter, the terminal **10** executes Auth and Config with the AP **6**. In Config with the AP **6**, the terminal **10** generates an SC for AP, and transmits the SC for AP to the AP **6**. Then, the terminal **10** and the AP **6** share a connection key by executing Network Access (hereinafter referred to as “NA”) using the SC for AP, and establish a Wi-Fi connection using the connection key. Thereafter, the terminal **10** executes BS, Auth, and Config (i.e., the steps T25 to T35 in FIG. 3) with the printer **100**. In Config with the printer **100**, the terminal **10** generates an SC for printer that does not include the SSID of the AP **6** and the password of the AP **6**, and transmits the SC for printer to the printer **100**. In this modification, the printer **100** and the AP **6** share the connection key by executing NA using the SC for printer and the SC for AP, and establish the Wi-Fi connection using the connection key. In this modification, the SC for printer is an example of “connection information”, “first connection information”, and “second connection information”.

#### Second Modification

(67) The Wi-Fi connection may be established between the terminal **10** and the printer **100**, instead of between the terminal **10** and the AP **6**. In this modification, the terminal **10** executes each processing T25 to T35 in FIG. 3, and transmits the SC for printer to the printer **100** in T35. The terminal **10** and the printer **100** share the connection key by executing NA using the SC for printer, and establish the Wi-Fi connection using the connection key. In this modification, the SC for printer is an example of “connection information”, “first connection information”, and “second connection information”. A pair of the terminal **10** and the printer **100** is an example of a “pair of apparatuses”.

#### Third Modification

(68) The Wi-Fi connection may be established between the printer **100** and the AP **6**, instead of between the terminal **10** and the AP **6**. In this modification, the terminal **10** executes each processing T25 to T35 in FIG. 3, and receives the SC for terminal from the printer **100** in T35. The terminal **10** and the AP **6** share the connection key by executing NA using the SC for terminal, and establish the Wi-Fi connection using the connection key. In this modification, a pair of the terminal **10** and the AP **6** is an example of a “pair of apparatuses”. Further, the “target communication” may include receiving “connection information”, “first connection information”, or “second connection information” from a communication apparatus. The “first target communication” may include receiving “connection information”, “first connection information”, or “second connection information” from a communication apparatus. The “second target communication” may include receiving “connection information”, “first connection information”, or “second connection information” from a communication apparatus.

#### Fourth Modification

(69) The “specific information” may be information such as a model name and a MAC address of the printer **100**, instead of a URI or a serial number.

#### Fifth Modification

(70) The character string **150** indicated by the QR Code may be described in accordance with the Bootstrapping Information Format, and may not be described according to the ABNF rule.

#### Sixth Modification

(71) The terminal **10A** may not include the connection application **41**. In this modification, the printer application **42** specifies the public key **170** in T122 of FIG. 4 and does not execute the step of T162. Then, the printer application **42** executes communication of T30 and T35 in FIG. 3 using the public key **170**. In this modification, executing communication of T30 and T35 using the public key **170** is an example of a “specific process”.

#### Seventh Modification

(72) The web page data that is transmitted from the server **200** to the terminal **10A** in T136 of FIG. 4 may include, for example, a character string and an image that constitute the web page **300**,

instead of the web page **300** itself. In this modification, the printer application **42** generates the web page **300** using the web page data.

#### Eighth Modification

(73) The communication system of the second embodiment may not include the server **200**. In this modification, the printer application **42** includes the model table **250**. Then, instead of T**324** to T**340** in FIG. **6**, the printer application **42** specifies a message corresponding to the serial number **162** from the model table **250**, and displays an operation screen including the message. In this modification, a process in which the printer application **42** specifies a message from the model table **250** and displays an operation screen is an example of a “display process”. In general, the “display process” may not include sending a web page request to the server and displaying a web page received from the server.

#### Ninth Modification

(74) The printer application **42** may provide the URI **160** specified at T**122** of FIG. **4** to the browser program. In this modification, the browser program can execute processing of T**130**, T**136**, and T**140**. The process in which the printer application **42** provides URI **160** to the browser program is an example of the “display process”.

#### Tenth Modification

(75) The “communication apparatus” may be another device such as a scanner, a multi-function device, a mobile terminal, a PC, or a server, instead of the printer **100**.

(76) In the above-described embodiment, each process in FIGS. **3** to **6** is realized by software, but at least one of these processes may be performed using hardware, such as a logic circuit.

(77) The technical elements described in this specification or the drawings each produce the technical effects separately or in various combinations, and are not limited to the combinations described in the claims as filed. In addition, the techniques illustrated in the specification and the drawings may achieve a plurality of objects at a time. Achieving one of the purposes may produce technical effects.

## Claims

1. A communication apparatus comprising: a wireless interface configured to execute a wireless communication according to Device Provisioning Protocol (DPP) scheme of Wi-Fi standard; a controller; a memory configured to store a computer-readable program; and a housing to which a seal is attached, the seal including a code image that is generated by encoding a character string, the character string being described in accordance with Bootstrapping Information Format and including a public key of the communication apparatus and specific information, the specific information being used for enabling a first terminal to execute a display process for displaying an operation screen of the communication apparatus, the operation screen indicating a changing operation for changing a state of the communication apparatus from a first state in which a wireless communication according to the DPP scheme is not to be executed to a second state in which the wireless communication according to the DPP scheme is to be executed; wherein the computer-readable program, when executed by the controller, causes the communication apparatus to: receive the changing operation for changing the state of the communication apparatus from the first state to the second state after the first terminal obtains the character string by decoding the code image of the seal; in a case where the changing operation is received, establish a wireless communication via the wireless interface with the first terminal according to the DPP scheme in response to the first terminal using the public key included in the character string.

2. The communication apparatus according to claim 1, wherein the character string includes: a first identifier according to Augmented Backus Naur Form (ABNF) rule; a first character string described in association with the first identifier and being the public key; a second identifier according to the ABNF rule; and a second character string described in association with the second

identifier and being the specific information.

3. The communication apparatus according to claim 1, wherein the display process includes: receiving, from a server, web page data for displaying the operation screen; and displaying, on the first terminal, the operation screen by using the received web page data.
  4. The communication apparatus according to claim 3, wherein the specific information is a Uniform Resource Identifier (URI) indicating a location of the web page data in the server.
  5. The communication apparatus according to claim 1, further comprising: a printer unit that prints an image on a sheet.
  6. The communication apparatus according to claim 1, further comprising: a scanner unit that scans an image on a sheet.
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